

Food & Beverage Analytics: Operational Dashboard

Transforming Raw Transactional Data into Strategic
Business Decisions.

Group 17 – Section A

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High-Volume Transactions Mask Critical Business Insights

Cafes generate massive amounts of daily data—item sales, timestamps, payment methods—but without structure, this information is noise rather than intelligence.

The Core Problem: How do item performance, payment preferences, and time-based sales trends impact overall revenue? Currently, decisions are made on intuition rather than evidence.

Project Objective



**Menu
Optimisation**



**Pricing
Strategy**



**Operational
Planning**

To engineer a performance dashboard that moves the stakeholder from reactive operations to proactive strategy.

Data Engineering: From 'Dirty' Raw Files to Analytical Assets



Raw Source

Kaggle Cafe Sales Dataset
10,000+ Rows
Unstructured CSV



Cleaning Pipeline

- **Null Handling:** Gap elimination for revenue accuracy
- **Standardisation:** Unified date formats for time-series
- **Deduplication:** Removal of inflated sales records
- **Text Normalisation:** Categorical grouping correction



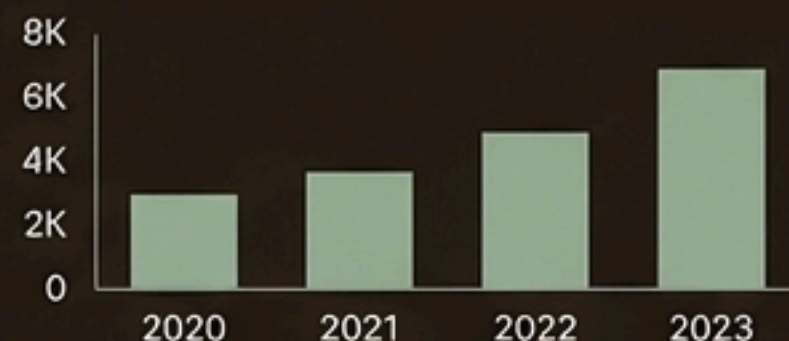
Clean Analytical Dataset

Validated Features: Transaction Date, Item Name, Quantity, Total Spent, Payment Method

A Framework for Measuring Commercial Success

We selected KPIs that link directly to revenue drivers and operational efficiency.

Total Revenue (Year-wise)



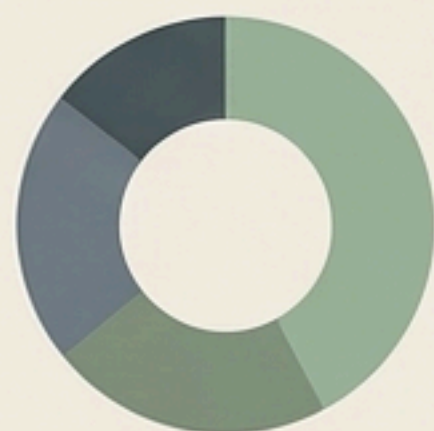
Top-line performance indicator.

Average Transaction Value (ATV)

\$20.97 ↗

Gauge for pricing strategy and upselling potential.

Revenue Share by Payment Method



Identifies checkout friction points.

Card Cash
Mobile App

Sales by Item (Volume vs Value)



Identifies menu winners and inventory draggers.

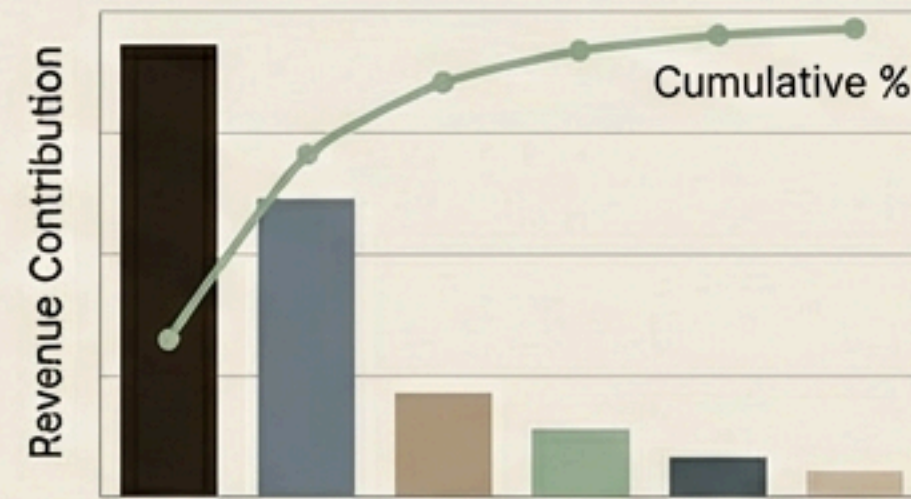
Mode of Shopping



Online vs In-store revenue streams.

Initial Discovery Reveals Distinct Consumption Patterns

The Pareto Principle



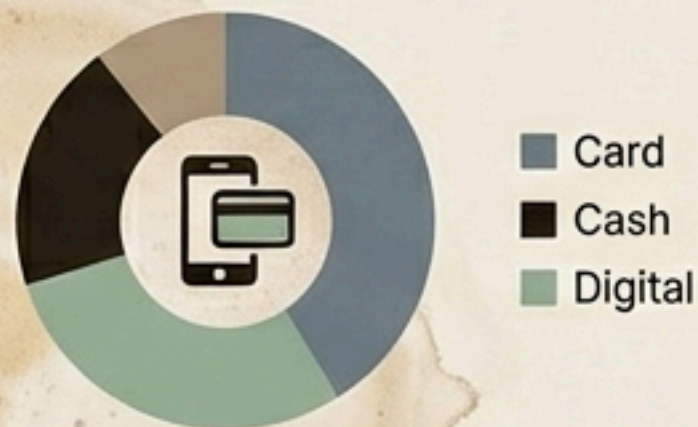
Small percentage of high-demand items drive majority of revenue.

Temporal Behaviour



Sales spike at specific hours and months, invalidating flat staffing models.

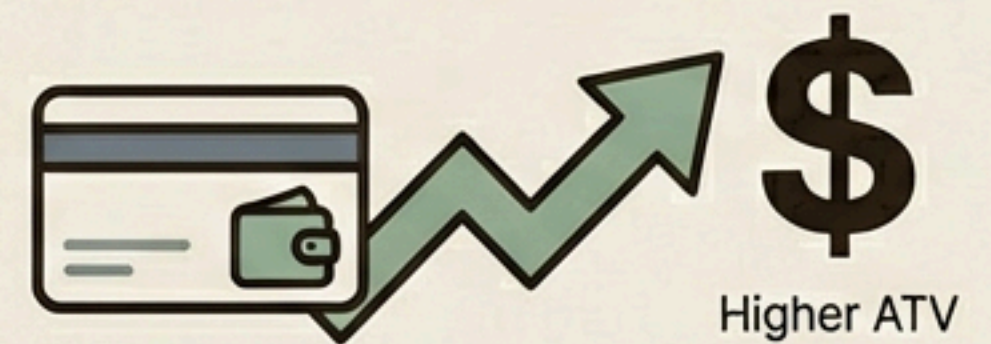
The Digital Shift



Digital payment methods dominate the revenue share.

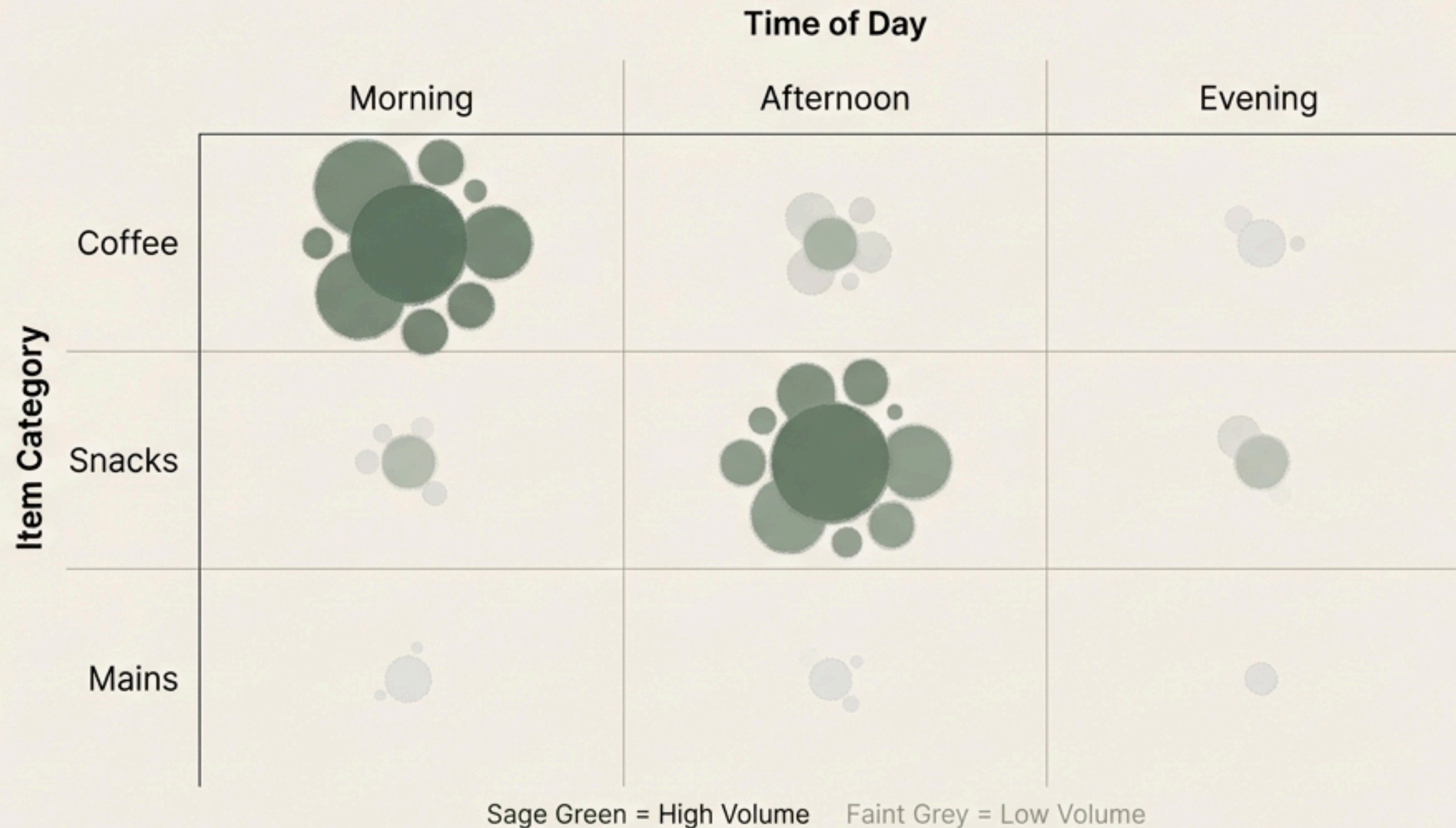
Value Correlation

Digital wallet users correlate with higher Average Transaction Values (ATV).



Advanced Analysis: Unlocking Efficiency through Segmentation

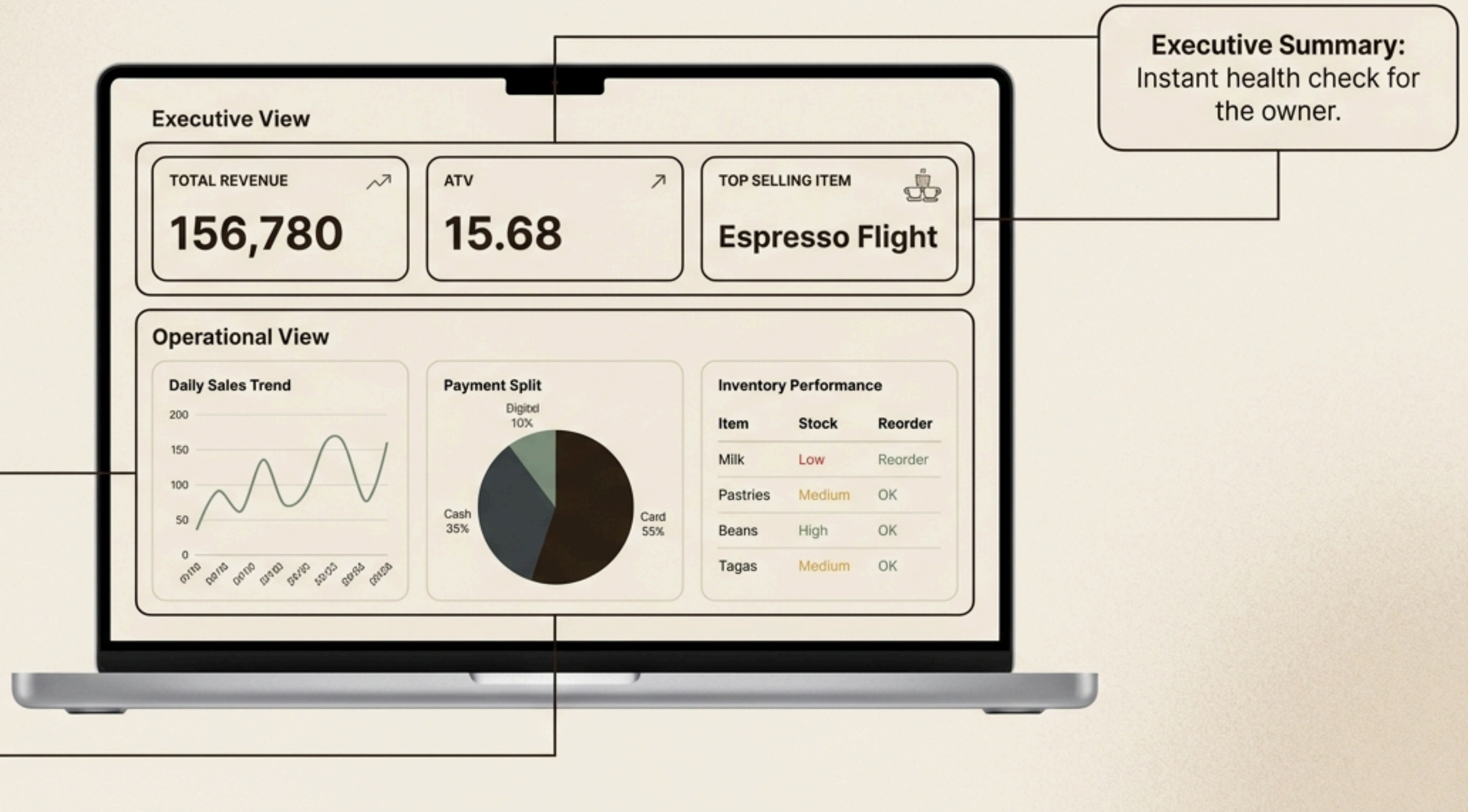
Correlation: Time of Day vs. Item Category



The New Understanding:

- 1. Predictable Bottlenecks:** Operations are not random; distinct consumption zones emerge.
- 2. Optimized Inventory:** Forecast specific needs (milk vs. pastries) by the hour.
- 3. Waste Reduction:** Move from 'stocked' to 'optimised' inventory levels.

The Dashboard: A Dual-View Operational Tool



Strategic Recommendations for Immediate Implementation

Menu Engineering



Inter

Identify bottom 20% of underperforming items. Remove or bundle to streamline inventory and reduce waste.

Dynamic Staffing



Inter

Align staff shifts with hourly/monthly sales peaks. Reduces idle labour costs and improves peak-hour service.

Checkout Incentives



Inter

Promote high-ATV payment methods (e.g., 5% off for Digital Wallet) to encourage higher spending and speed.

Business Impact: Efficiency, Revenue, and Confidence

1. Revenue Uplift

Direct top-line growth through optimized menu pricing and focus on high-ATV items.

2. Operational Efficiency

Significant reduction in food waste and stockouts via time-series inventory forecasting.

3. Stakeholder Value

Replaces gut feeling with data-backed confidence. Provides a scalable framework for future location expansion.

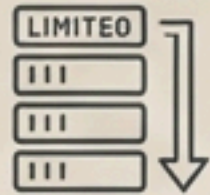


🌱 *Transforming raw operational data into a competitive advantage.* 🌱

Project Limitations & The Path Forward

Current Limitations

CURRENT
STATE



- **Dataset Constraints:** Kaggle source (~10k rows) is a training set; limited multi-year seasonality.



- **Scope:** Retrospective analysis rather than real-time monitoring.

Future Roadmap

FUTURE
STATE



- **Integration:** Connect dashboard to live POS API feeds.



- **Predictive Modelling:** Machine learning for 'Next Week' sales forecasting.



- **Customer Loyalty:** Integrate Customer ID data for Lifetime Value (CLV) tracking.

Thankyou